SAT Dashboard Methodology

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I. Purpose and Scope

This document describes the methodology used in the Sustainable Aviation Technology Dashboard (SAT Dashboard). The SAT Dashboard provides a screening level scenario analysis framework for comparing alternative aviation energy pathways, including battery electric propulsion, sustainable aviation fuel (SAF), hydrogen propulsion, and a parametric future electrochemical cell technology module referred to as Energy eXploration, or Energy X. The tool is intended to support research, education, and early stage policy and technology exploration by making key assumptions, input data, and derived quantities transparent and accessible.

The SAT Dashboard is not intended to replace detailed aircraft design tools, certification analyses, techno economic assessment packages, airport infrastructure models, or pathway specific life cycle assessment software. Instead, it provides first order estimates of route feasibility, passenger allocation, energy source cost per passenger mile, fuel use, land use requirements, and greenhouse gas emissions under user defined assumptions.

The methodology presented here is intended to make the dashboard outputs traceable and reproducible. For each major module, this document identifies the governing equations, defines the relevant variables and describes the role of the input data.

Accordingly, the models implemented in the dashboard should be interpreted as transparent screening models. Their outputs are most appropriate for evaluating trends, sensitivities, and relative differences among energy pathways, rather than for making final design, certification, policy, or investment decisions.

II. Data Sources and Traceability

The SAT Dashboard uses three main categories of data: direct input data, internally derived quantities, and auxiliary reference properties. Direct input data are values obtained from public datasets, government databases, manufacturer datasheets, and published literature. These values are used directly in the dashboard calculations. Examples include route distance, passenger demand, monthly flight counts, fuel consumed per flight, total fuel per route, fuel cost, estimated aircraft capacity, battery specific energy, battery maximum C-rate, hydrogen production emissions factors, SAF life-cycle emission factors, and SAF feedstock yield values. Route-level operations data, including passenger demand, route distance, monthly flight counts, and fuel-related quantities, are based on airline operations data from the Bureau of Transportation Statistics [1]. SAF pathway properties and feedstock conversion assumptions are based on published SAF pathway datasets and CORSIA-compatible fuel documentation [2, 3]. Life-cycle greenhouse-gas emission factors are treated consistently with GREET-style life-cycle assessment methods [4].

Internally derived quantities are calculated within the dashboard using the equations documented in this methodology. These include electric aircraft range, battery pack charge, the number of cells in series and parallel, current feasibility, passenger displacement, hydrogen aircraft range, hydrogen passenger displacement, SAF land-use requirements, energy-source cost per passenger-mile, and cumulative fleet emissions.

Auxiliary reference properties are included to provide context for the user but are not used directly in route feasibility, cost, or emissions calculations unless explicitly stated. Examples include fuel freezing point, flash point, and selected compatibility-related fuel properties.

Table 1 summarizes the main quantities used in the dashboard, their role, their classification, and their typical source.

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Table 1 Traceability structure for key quantities used in the SAT Dashboard.

Quantity	Role in Dashboard	Type	Typical Source
Route distance	Determines route feasibility and passenger-mile totals	Direct input	Flight operations data [1]
Passenger demand	Defines addressable demand and passenger displacement	Direct input	Flight operations data [1]
Monthly flight count	Scales monthly passengers, fuel use, and emissions	Direct input	Flight operations data [1]
Estimated aircraft capacity	Defines aircraft class and representative parameters	Direct input	Flight operations data and aircraft database
Fuel consumed per flight	Defines conventional fuel use and emissions	Direct input	Flight operations data [1]
Total fuel per route	Defines monthly route fuel demand and emissions	Direct input	Flight operations data [1]
Fuel cost	Defines conventional energy-source cost	Direct input	Flight operations data [1]
Battery specific energy	Determines stored energy and electric range	Direct input	Manufacturer datasheets and literature
Battery maximum C-rate	Determines current-delivery feasibility	Direct input	Manufacturer datasheets and literature
Battery operating temperature	Determines thermal route feasibility	Direct input	Manufacturer datasheets and literature
SAF maximum blend ratio	Limits the effective neat SAF fraction	Direct input	SAF dataset and CORSIA-related sources [2]
SAF life-cycle emission factor	Computes SAF life-cycle emissions	Direct input	GREET, CORSIA, and literature [2, 4]
SAF volumetric energy density	Converts SAF volume to energy	Direct input or derived	Fuel property data
SAF yield per acre	Estimates feedstock land-use requirements	Direct input or derived	USDA yield data and conversion factors [2, 3]
Hydrogen GHG emission factor	Computes hydrogen pathway emissions	Direct input	Hydrogen dataset and GREET-style LCA [4]
Hydrogen production pathway shares	Defines the weighted hydrogen emissions factor	User input	Dashboard input
Electricity cost	Computes electric energy-source cost	User input	Dashboard input
Fuel cost per gallon	Computes SAF and hydrogen energy-source cost	User input	Dashboard input

III. Range Estimation of Alternative Energy Carriers

Electrification

Battery-electric propulsion introduces range, payload, and power-delivery constraints that are not present for drop-in fuels such as SAF. The Electrification module evaluates whether a route can be served by a representative battery-electric aircraft under user-defined assumptions for battery technology, battery mass fraction, aircraft system voltage, propulsive efficiency, operating month, airline adoption, and fleet adoption. Route distance, passenger demand, monthly operations, and related flight quantities are obtained the 2019 U.S. Bureau of Transportation Statistics (BTS) [1]

Representative aircraft parameters are assigned according to the estimated aircraft capacity reported in the flight-operations dataset. Each capacity group is mapped to a representative maximum power, takeoff gross weight, and lift-to-drag ratio, as summarized in Table 2. These parameters provide a screening-level aircraft representation for the Electrification and Energy-X modules.

Table 2 Representative aircraft parameters used in the Electrification and Energy-X modules.

Estimated Capacity	P_{max} [W]	L/D	TOGW [kg]	Representative Class
19	1,158,000	14	6,575	Commuter
88	2,050,000	15	23,000	Regional
120	13,567,500	16	63,100	Short-haul
189	15,270,000	18	79,015.8	Medium-haul
368	68,000,000	19	227,900	Large transport

The Electrification module estimates route feasibility using a constant-mass form of the electric Breguet range relation. This formulation provides a first-order estimate of the maximum range achievable by a representative battery-electric aircraft under the user-selected technology assumptions:

$$R = \eta_p \frac{L}{D} \frac{e_{\text{cell}}}{g} f_{\text{bat}}, \tag{1}$$

where R is the estimated aircraft range in meters, η_p is the user-defined propulsion-system efficiency, L/D is the representative lift-to-drag ratio, e_{cell} is the battery cell specific energy in J/kg, g is the gravitational acceleration, and f_{bat} is the user-selected battery mass fraction. In the dashboard implementation, the electrical, mechanical, and propulsive efficiency effects are represented through the single efficiency parameter η_p. The cell specific energy entered by the user in Wh/kg is converted to J/kg according to

$$e_{\text{cell}} = 3.6 \times 10^3 e_{\text{cell,Wh/kg}}, \tag{2}$$

where $e_{\text{cell,Wh/kg}}$ is the cell specific energy in Wh/kg. The estimated range is then converted from meters to miles using

$$R_{\text{mi}} = 6.21371 \times 10^{-4} R, \tag{3}$$

where R_{mi} is the estimated electric aircraft range in miles. A route is classified as range feasible if the estimated electric range exceeds the route distance,

$$D_{\text{route}} < R_{\text{mi}}, \tag{4}$$

where D_{route} is the route distance in miles.

The selected battery mass fraction also determines the onboard battery mass. For an aircraft with takeoff mass m_{TO}, the battery mass is computed as

$$m_{\text{bat}} = f_{\text{bat}} m_{\text{TO}}, \tag{5}$$

where m_{bat} is the battery mass. The corresponding stored battery energy is

$$E_{\text{bat}} = m_{\text{bat}} e_{\text{cell}}, \tag{6}$$

where E_{bat} is the total stored battery energy in joules.

In addition to the range constraint, the dashboard evaluates whether the selected battery configuration can satisfy the maximum aircraft power requirement. The user specifies the battery bus voltage V_{bat}, and the required battery current is calculated from the representative maximum aircraft power $P_{\max}$ as

$$I_{\text{bat}} = \frac{P_{\max}}{V_{\text{bat}}}, \tag{7}$$

where I_{bat} is the total bus current. The number of cells in series is estimated from the ratio between the battery bus voltage and the nominal cell voltage,

$$n_s = \frac{V_{\text{bat}}}{V_{\text{cell}}}, \tag{8}$$

where n_s is the number of cells in series and V_{cell} is the nominal cell voltage. The total battery charge is obtained from the stored energy and bus voltage,

$$Q_{\text{bat}} = \frac{E_{\text{bat}}}{V_{\text{bat}}}, \tag{9}$$

where Q_{bat} is the total battery charge in coulombs. The charge capacity of one cell is computed from the selected cell capacity C_{Ah} as

$$Q_{\text{cell}} = 3.6 \times 10^3 C_{\text{Ah}}, \tag{10}$$

where Q_{cell} is the charge capacity of one cell in coulombs and C_{Ah} is the cell capacity in ampere hours. The number of parallel cells required to provide the stored charge is then estimated as

$$n_p = \frac{Q_{\text{bat}}}{Q_{\text{cell}}}, \tag{11}$$

where n_p is the number of cells in parallel implied by the stored-energy requirement.

The current-delivery capability of the pack is evaluated using the maximum cell C-rate. The maximum allowable current per cell is

$$I_{\text{cell,max}} = C_{\text{Ah}} C_{\max}, \tag{12}$$

where $I_{\text{cell,max}}$ is the maximum allowable cell current and $C_{\max}$ is the maximum C-rate. The minimum number of parallel cells required to satisfy the aircraft current demand is

$$n_{p,\min} = \frac{I_{\text{bat}}}{I_{\text{cell,max}}}, \tag{13}$$

where $n_{p,\min}$ is the minimum number of parallel cells required by the current constraint. The battery configuration is classified as current feasible when

$$n_p \geq n_{p,\min}. \tag{14}$$

If Eq. (14) is not satisfied, the route is treated as infeasible in the current implementation, and its electric range is set to zero.

The module also accounts for passenger displacement caused by the difference between the battery mass and the conventional Jet-A fuel mass associated with the route. The conventional fuel mass is computed from the route fuel volume and Jet-A density as

$$m_f = \rho_{\text{JetA}} V_f, \tag{15}$$

where m_f is the conventional fuel mass, ρ_{JetA} is the Jet-A density, and V_f is the route fuel volume. The residual mass penalty associated with replacing the conventional fuel with batteries is

$$m_{\text{res}} = m_{\text{bat}} - m_f, \tag{16}$$

where m_{res} is the additional mass that must be offset by removing payload. The number of displaced passengers is estimated as

$$\Delta N_{\text{pax}} = \left\lceil \frac{m_{\text{res}}}{m_{\text{pax}}} \right\rceil, \tag{17}$$

where ΔN_{pax} is the number of passengers removed and m_{pax} is the assumed passenger plus luggage mass. Negative values of ΔN_{pax} are set to zero. The current implementation uses

$$m_{\text{pax}} = 158.757 \text{ kg}. \tag{18}$$

The resulting monthly passenger volume assigned to electric operation on a route is

$$N_{\text{pax,E}} = N_{\text{pax,route}} - \Delta N_{\text{pax}} N_{\text{flights}}, \tag{19}$$

where $N_{\text{pax,E}}$ is the remaining electric passenger volume, $N_{\text{pax,route}}$ is the original monthly passenger demand on the route, and N_{flights} is the number of monthly flights. Routes with non-positive values of $N_{\text{pax,E}}$ are classified as infeasible.

Thermal feasibility is evaluated using the selected battery operating-temperature limits and the monthly average temperatures at the origin and destination airports. A route remains feasible only when both endpoint temperatures lie within the allowable battery operating range,

$$T_{\min} < T_{\text{origin,month}} < T_{\max}, \tag{20}$$

and

$$T_{\min} < T_{\text{destination,month}} < T_{\max}, \tag{21}$$

where $T_{\min}$ and $T_{\max}$ are the minimum and maximum allowable battery operating temperatures, and $T_{\text{origin,month}}$ and $T_{\text{destination,month}}$ are the monthly average temperatures at the route origin and destination, respectively.

The full route-screening process is illustrated in Figure 1. The dashboard first evaluates battery current capability and range feasibility, then applies passenger displacement, airline selection, aircraft-class selection, temperature, and adoption filters. The resulting feasible route set is used to generate route maps, passenger-distance distributions, busiest-airport rankings, passenger-mile shares, energy-source cost metrics, and monthly emissions estimates.

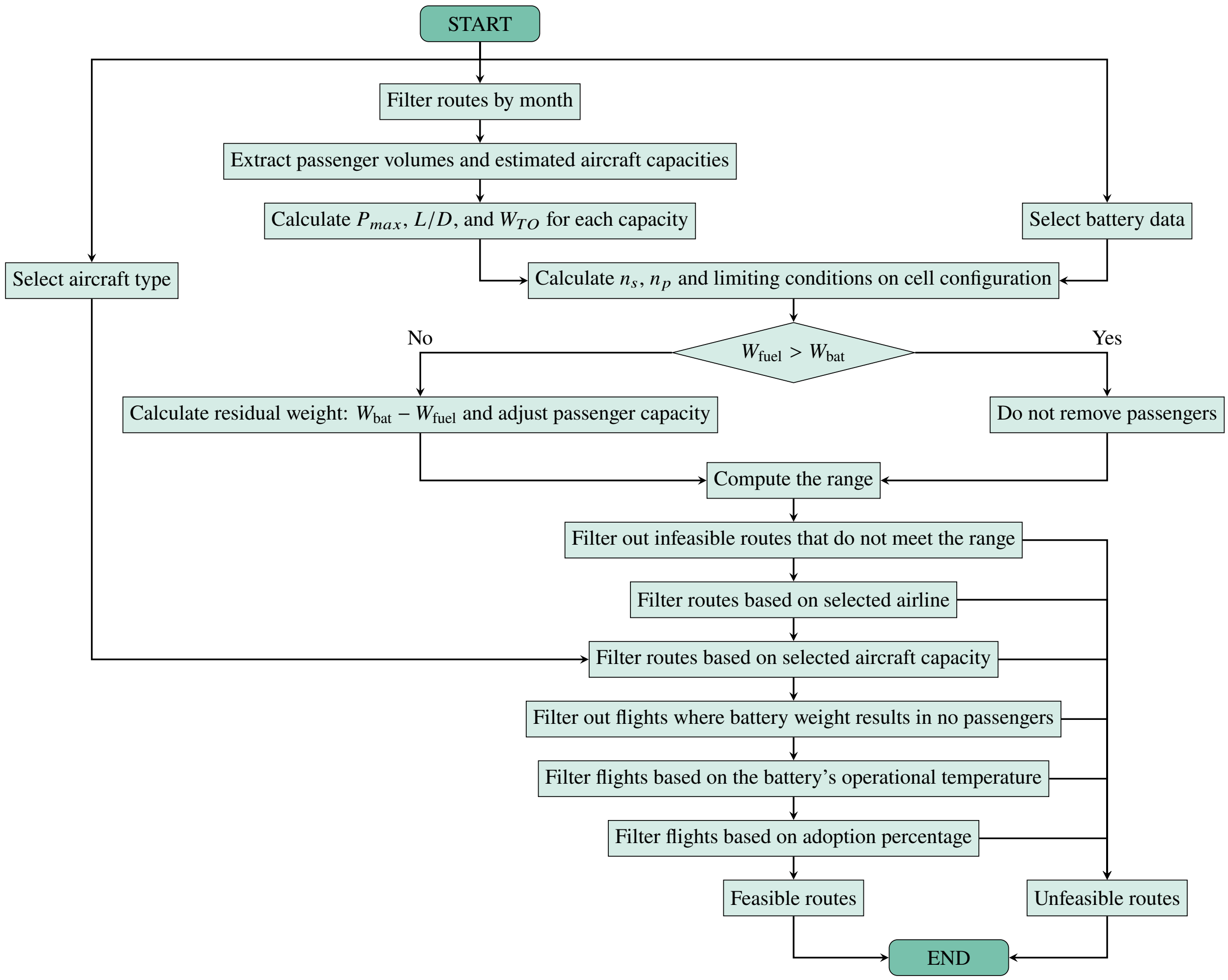

Fig. 1 Flowchart of electric route feasibility evaluation implemented in the SAT Dashboard.

Energy-eX(ploration)

The Energy-eX(ploration), or Energy-X, module follows the same route feasibility framework used in the Electrification module. However, rather than selecting a battery technology from the dashboard database, the user defines a hypothetical future cell by specifying its nominal voltage, capacity, maximum C-rate, specific energy, and allowable operating temperature range. This structure allows the module to evaluate how improvements in cell-level performance could affect aircraft range, passenger allocation, and route feasibility across the domestic aviation network.

The user-defined system voltage is entered in kilovolts and converted internally to volts according to

$$V_{\text{bat}} = 1.0 \times 10^3 V_{\text{system,kV}}, \tag{22}$$

where $V_{\text{system,kV}}$ is the selected system voltage in kV and V_{bat} is the corresponding battery bus voltage in V. The cell specific energy is converted from Wh/kg to J/kg using

$$e_{\text{cell}} = 3.6 \times 10^3 e_{\text{cell,Wh/kg}}, \tag{23}$$

where $e_{\text{cell,Wh/kg}}$ is the user-defined cell specific energy in Wh/kg and e_{cell} is the value used in the range and battery energy calculations.

The cell charge is computed from the user-specified cell capacity as

$$Q_{\text{cell}} = 3.6 \times 10^3 C_{\text{Ah}}, \tag{24}$$

where C_{Ah} is the cell capacity in ampere hours and Q_{cell} is the cell charge in coulombs. The maximum allowable current per cell is estimated from the selected maximum C-rate:

$$I_{\text{cell,max}} = C_{\text{Ah}} C_{\text{max}}, \tag{25}$$

where C_{max} is the maximum discharge rate and $I_{\text{cell,max}}$ is the maximum allowable cell current.

After these cell properties are defined, the Energy-X module applies the same battery pack sizing, current feasibility, range estimation, passenger displacement, temperature feasibility, and adoption filters described for the Electrification module. The quantities I_{bat}, n_s, m_{bat}, E_{bat}, Q_{bat}, n_p, $n_{p,\text{min}}$, and the resulting route feasibility classification are therefore computed using the same governing relationships. Consequently, Energy-X should be interpreted as a parametric sensitivity module for future electrochemical cell performance, rather than as a separate aircraft sizing or optimization model.

Sustainable Aviation Fuel

The Sustainable Aviation Fuel, or SAF, module evaluates how fuel selection, blend limits, airport availability, feedstock availability, and adoption assumptions influence route allocation, land use requirements, energy source cost, and emissions. In contrast to the Electrification and Hydrogen modules, SAF operation is not assigned a range penalty or passenger displacement penalty. This reflects the dashboard assumption that SAF is used as a drop in, or near drop in, replacement for conventional Jet-A, subject to maximum blend ratio constraints and airport availability.

The feedstock and production pathway treatment follows prior feedstock to fuel modeling work [3]. The alcohol to jet, or ATJ, and hydroprocessed esters and fatty acids, or HEFA, conversion assumptions are based on CORSIA compatible SAF pathway documentation [2]. Corn and sorghum are treated with similar grain to ethanol conversion behavior because of their comparable starch content characteristics [5].

Users select one or more Jet-A and SAF fuel options from the commercial SAF dataset and define the intended fuel use distribution with a range slider. The selected fuels are then combined with their maximum allowable blend ratios to determine the effective neat SAF fraction in the fuel mixture. If the conventional Jet-A entry is not selected explicitly, it is automatically included so that the remaining fuel fraction can be assigned to Jet-A.

The slider values are interpreted as cumulative fuel use breakpoints. If p_i denotes the cumulative breakpoint associated with fuel i, the nominal share assigned to that fuel is

$$x_i = \frac{p_i - p_{i-1}}{100}, \tag{26}$$

where x_i is the fractional fuel share and p_i is expressed as a percentage. For SAF fuels, this nominal share is limited by the maximum allowable blend ratio,

$$\phi_i = x_i \frac{BR_i}{100}, \tag{27}$$

where ϕ_i is the effective neat SAF fraction assigned to fuel i, and BR_i is the maximum blend ratio of that fuel in percent. The remaining fuel fraction is assigned to conventional Jet-A:

$$\phi_{\text{JetA}} = 1 - \sum_{i=1}^{N_{\text{SAF}}} \phi_i, \tag{28}$$

where N_{SAF} is the number of selected SAF fuels.

Airport adoption is modeled through an origin airport availability filter. The user specifies whether SAF is available at the top 5, top 10, top 20, top 50, or all origin airports, ranked by passenger volume. Routes departing from airports outside the selected availability set remain assigned to the conventional Jet-A fleet. Routes departing from SAF enabled airports are then sampled according to the user defined fleet adoption percentage,

$$\mathcal{R}_{\text{SAF}} = \text{sample}\left(\mathcal{R}_{\text{SAF,airports}}, \frac{p_{\text{adopt}}}{100}\right), \tag{29}$$

where $\mathcal{R}_{\text{SAF}}$ is the set of routes assigned to SAF operation, $\mathcal{R}_{\text{SAF,airports}}$ is the set of routes departing from SAF enabled airports, and p_{adopt} is the selected adoption percentage. Routes that are eligible for SAF but not selected during the sampling step remain assigned to Jet-A. The SAF module therefore represents a user defined deployment scenario rather than an optimized allocation of SAF across the network.

For routes assigned to SAF operation, the total fuel volume is computed as

$$V_{\text{fuel,total}} = \sum_{r \in \mathcal{R}_{\text{SAF}}} V_{\text{fuel},r}, \tag{30}$$

where $V_{\text{fuel},r}$ is the fuel volume associated with route r. The volume assigned to each selected fuel is then

$$V_i = \phi_i V_{\text{fuel,total}}, \tag{31}$$

where V_i is the volume assigned to fuel i.

Land use requirements are estimated from the volume of SAF that can be produced per unit land area. The SAF yield is calculated as

$$Y_{\text{SAF}} = Y_{\text{crop}} F_{\text{feedstock}} F_{\text{residue}} F_{\text{SAF}}, \tag{32}$$

where Y_{SAF} is the SAF yield in gal/acre, Y_{crop} is the crop yield in bu/acre, $F_{\text{feedstock}}$ is the feedstock mass per bushel in kg/bu, F_{residue} is the residue to feedstock mass ratio in $\text{kg}_{\text{residue}}/\text{kg}_{\text{feedstock}}$, and F_{SAF} is the SAF yield per unit residue mass in $\text{gal}_{\text{SAF}}/\text{kg}_{\text{residue}}$. The land area required to supply a specified SAF volume is then

$$A_{\text{land}} = \frac{V_{\text{SAF,required}}}{Y_{\text{SAF}}}, \tag{33}$$

where A_{land} is the required land area in acres and $V_{\text{SAF,required}}$ is the required SAF volume in gallons.

The dashboard aggregates the required SAF volume by feedstock type and estimates the corresponding acreage requirement. The user selects a feedstock crop and the states from which that feedstock may be sourced. County level crop records are then randomly ordered, and harvested acreage is accumulated until the required acreage is satisfied. The counties used to meet the feedstock requirement are highlighted in the land use map. This procedure provides a first order estimate of land availability and geographic distribution, but it does not represent a supply chain optimization, economic allocation model, or crop displacement analysis.

The SAF route allocation workflow is summarized in Figure 2. After the user selects fuels, blend assumptions, airport availability, and adoption levels, the dashboard applies the route eligibility and adoption filters to determine which routes are assigned to SAF operation and which routes remain assigned to conventional Jet-A.

Hydrogen

The Hydrogen module provides a screening level estimate of the operational feasibility of hydrogen powered aircraft across a route network. The module identifies routes that could be assigned to hydrogen operation under user defined assumptions for hydrogen storage volume, cruise altitude, hydrogen engine specific fuel consumption, hydrogen production pathway, airport infrastructure availability, fleet adoption, and hydrogen cost. The model is intended for scenario analysis and technology comparison. It does not represent a detailed aircraft retrofit model, cryogenic tank design, propulsion cycle analysis, or certification level assessment.

The atmospheric properties used in the hydrogen range calculation are estimated at the user selected cruise altitude. The geometric altitude, h_{geo}, is first converted to geopotential altitude according to

$$h = \frac{r_e}{r_e + h_{\text{geo}}} h_{\text{geo}}, \tag{34}$$

where h is the geopotential altitude and r_e is the mean Earth radius. For altitudes below approximately 3.6×10^4 ft, the temperature is computed using a linear lapse rate relation,

$$T = T_0 + a_1 (h - h_0), \tag{35}$$

where T is the atmospheric temperature, T_0 is the reference temperature, a_1 is the lapse rate, and h_0 is the reference altitude. The corresponding density is estimated as

$$\rho = \rho_0 \left(\frac{T}{T_0}\right)^{-\left(\frac{g_0}{R a_1}+1\right)}, \tag{36}$$

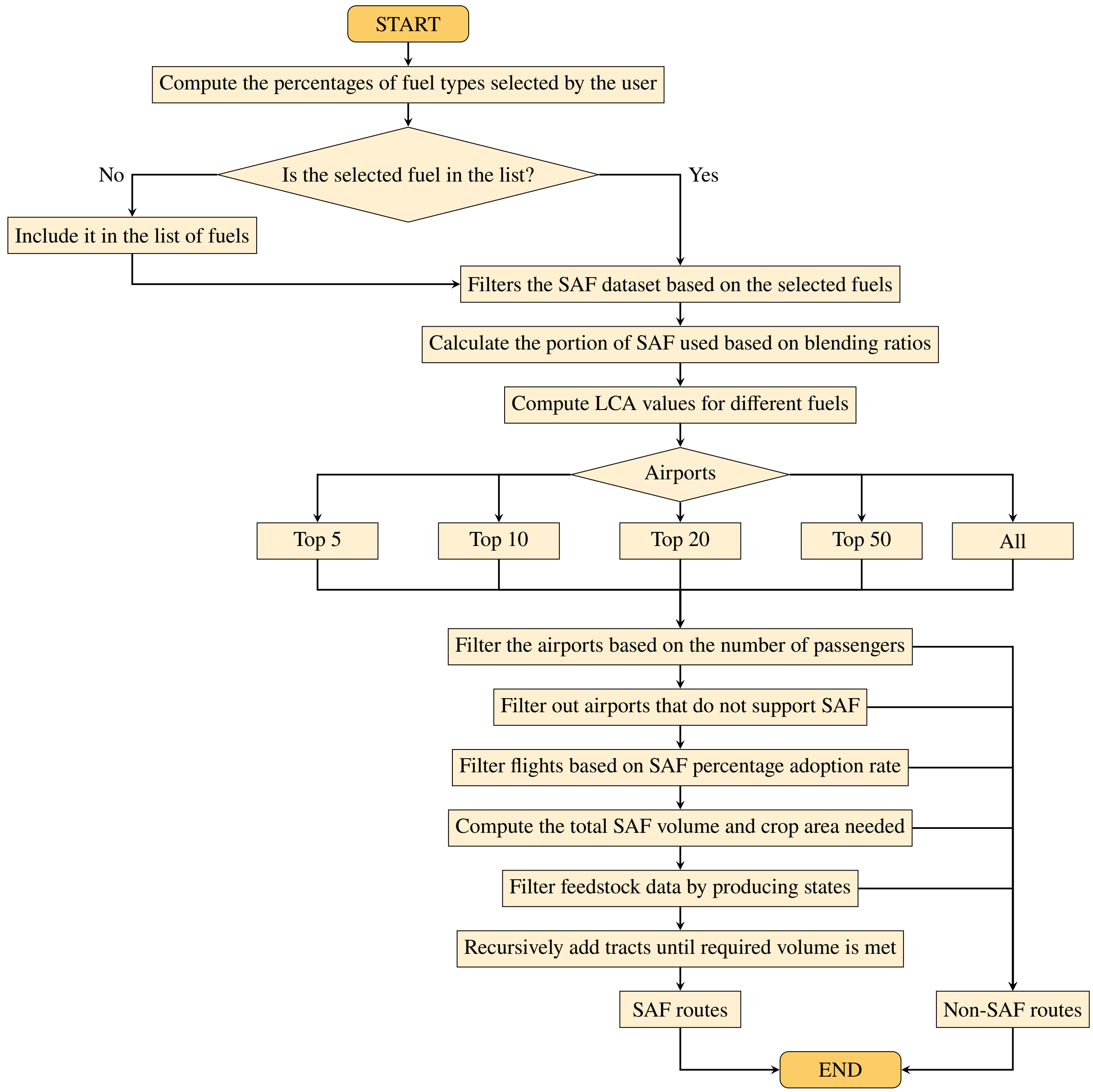

Fig. 2 Flowchart of the SAF route selection process implemented in the dashboard.

where ρ is the atmospheric density, ρ_0 is the reference density, g_0 is the gravitational acceleration, and R is the specific gas constant for air. For altitudes above approximately 3.6×10^4 ft, the implementation uses an exponential density relation,

$$\rho = \rho_0 \exp\left[-\frac{g_0}{RT}(h - h_0)\right]. \tag{37}$$

The resulting density is converted to SI units before being used in the range calculation.

Representative aircraft parameters are estimated as functions of the aircraft passenger capacity, C. The available aircraft volume is approximated by the polynomial relation

$$V_{\text{aircraft}} = 3.0 \times 10^{-4} C^3 - 8.93 \times 10^{-2} C^2 + 10.1C - 314.21, \tag{38}$$

where V_{aircraft} is the estimated available aircraft volume. The empty weight is estimated as

$$W_{\text{empty}} = \left(1.2636C^2 + 145.58C\right) g, \tag{39}$$

where W_{empty} is the aircraft empty weight and g is the gravitational acceleration. The representative wing area is calculated as

$$S = 5.9 \times 10^{-3}C^2 - 9.942 \times 10^{-1}C + 132.03, \tag{40}$$

where S is the wing reference area. The representative cruise lift and drag coefficients are estimated as

$$C_L = -6.0 \times 10^{-6}C^2 + 2.8 \times 10^{-3}C + 2.695 \times 10^{-1}, \tag{41}$$

and

$$C_D = 8.0 \times 10^{-7}C^2 - 3.0 \times 10^{-4}C + 6.15 \times 10^{-2}, \tag{42}$$

where C_L and C_D are the representative cruise lift and drag coefficients.

Hydrogen storage is modeled by assigning a user selected fraction of the available aircraft volume to liquid hydrogen storage. The selected percentage is converted to a volume fraction as

$$f_{V,H_2} = \frac{V_{H_2,\%}}{100}, \tag{43}$$

where f_{V,H_2} is the hydrogen storage volume fraction and $V_{H_2,\%}$ is the user selected hydrogen storage percentage. The corresponding hydrogen storage volume is

$$V_{H_2} = f_{V,H_2} V_{\text{aircraft}}, \tag{44}$$

where V_{H_2} is the hydrogen storage volume. The hydrogen weight is computed from a fixed liquid hydrogen density,

$$W_{H_2} = V_{H_2} \rho_{H_2} g, \tag{45}$$

where W_{H_2} is the hydrogen weight and ρ_{H_2} is the assumed liquid hydrogen density. The current implementation uses

$$\rho_{H_2} = 70 \text{ kg/m}^3. \tag{46}$$

Passenger weight is reduced according to the fraction of aircraft volume assigned to hydrogen storage:

$$W_{\text{pax}} = \left(1 - f_{V,H_2}\right) m_{\text{pax}} g C, \tag{47}$$

where W_{pax} is the passenger weight and m_{pax} is the assumed passenger plus luggage mass. In the current implementation,

$$m_{\text{pax}} = 250 \text{ kg}. \tag{48}$$

This value should be interpreted as a conservative implementation assumption. A future revision may replace it with a direct SI conversion of a 250 lb passenger plus luggage assumption.

The initial and final aircraft weights used in the hydrogen range calculation are

$$W_0 = W_{\text{empty}} + W_{\text{pax}} + W_{H_2}, \tag{49}$$

and

$$W_f = W_0 - 0.95 W_{H_2}, \tag{50}$$

where W_0 is the initial aircraft weight and W_f is the final aircraft weight. Equation (50) assumes that 95% of the onboard hydrogen mass is usable during the mission.

The hydrogen engine specific fuel consumption is supplied by the user in imperial units and converted internally to SI units as

$$\mathrm{SFC}_{H_2,\mathrm{SI}} = 2.8325 \times 10^{-5} \mathrm{SFC}_{H_2,\mathrm{Imperial}}, \tag{51}$$

where $\mathrm{SFC}_{H_2,\mathrm{Imperial}}$ is the user supplied hydrogen specific fuel consumption and $\mathrm{SFC}_{H_2,\mathrm{SI}}$ is the value used in the range calculation.

The hydrogen aircraft range is then estimated as

$$R = \frac{1}{\mathrm{SFC}_{H_2,\mathrm{SI}}} \frac{C_L^2}{C_D} \sqrt{\frac{2}{\rho S}} \left(\sqrt{W_0} - \sqrt{W_f} \right), \tag{52}$$

where R is the estimated range in meters. The range is converted from meters to miles using

$$R_{\mathrm{mi}} = 6.21371 \times 10^{-4} R, \tag{53}$$

where R_{mi} is the estimated hydrogen aircraft range in miles. A route is classified as range feasible when

$$D_{\mathrm{route}} < R_{\mathrm{mi}}, \tag{54}$$

where D_{route} is the route distance in miles.

Passenger feasibility is estimated from the route level passenger utilization. The passenger utilization factor is

$$\phi_{\mathrm{pax}} = \frac{N_{\mathrm{pax,route}}}{N_{\mathrm{flights}} C}, \tag{55}$$

where $N_{\mathrm{pax,route}}$ is the monthly passenger demand on the route and N_{flights} is the number of monthly flights. The unused aircraft volume fraction is approximated as

$$\phi_{\mathrm{unused}} = 1 - \phi_{\mathrm{pax}}, \tag{56}$$

where ϕ_{unused} represents the fraction of nominal passenger capacity not used by the route. The additional hydrogen volume fraction that cannot be absorbed by unused passenger volume is estimated as

$$\phi_{\mathrm{add}} = \phi_{\mathrm{unused}} - f_{V,H_2}. \tag{57}$$

Negative values of ϕ_{add} are set to zero in the implementation. The remaining hydrogen passenger count is then approximated as

$$N_{\mathrm{pax},H_2} = N_{\mathrm{pax,route}} - (1 + \phi_{\mathrm{add}})\, C, \tag{58}$$

where N_{pax,H_2} is the passenger count remaining after the hydrogen volume penalty is applied. Routes with non positive values of N_{pax,H_2} are classified as passenger infeasible.

Hydrogen airport availability is modeled using an origin airport infrastructure filter. The user specifies whether hydrogen is available at the top 5, top 10, top 20, top 50, or all origin airports, ranked by passenger volume. Routes departing from airports outside the selected set remain assigned to the non hydrogen fleet. Routes departing from hydrogen enabled airports are filtered by range feasibility and passenger feasibility. The final set of hydrogen operated routes is then selected by sampling the user defined adoption fraction from the eligible hydrogen route set:

$$\mathcal{R}_{H_2} = \mathrm{sample}\left(\mathcal{R}_{\mathrm{eligible}}, \frac{p_{\mathrm{adopt}}}{100} \right), \tag{59}$$

where $\mathcal{R}_{H_2}$ is the set of routes assigned to hydrogen operation, $\mathcal{R}_{\mathrm{eligible}}$ is the set of hydrogen eligible routes, and p_{adopt} is the selected adoption percentage. The resulting hydrogen route allocation workflow is summarized in Figure 3.

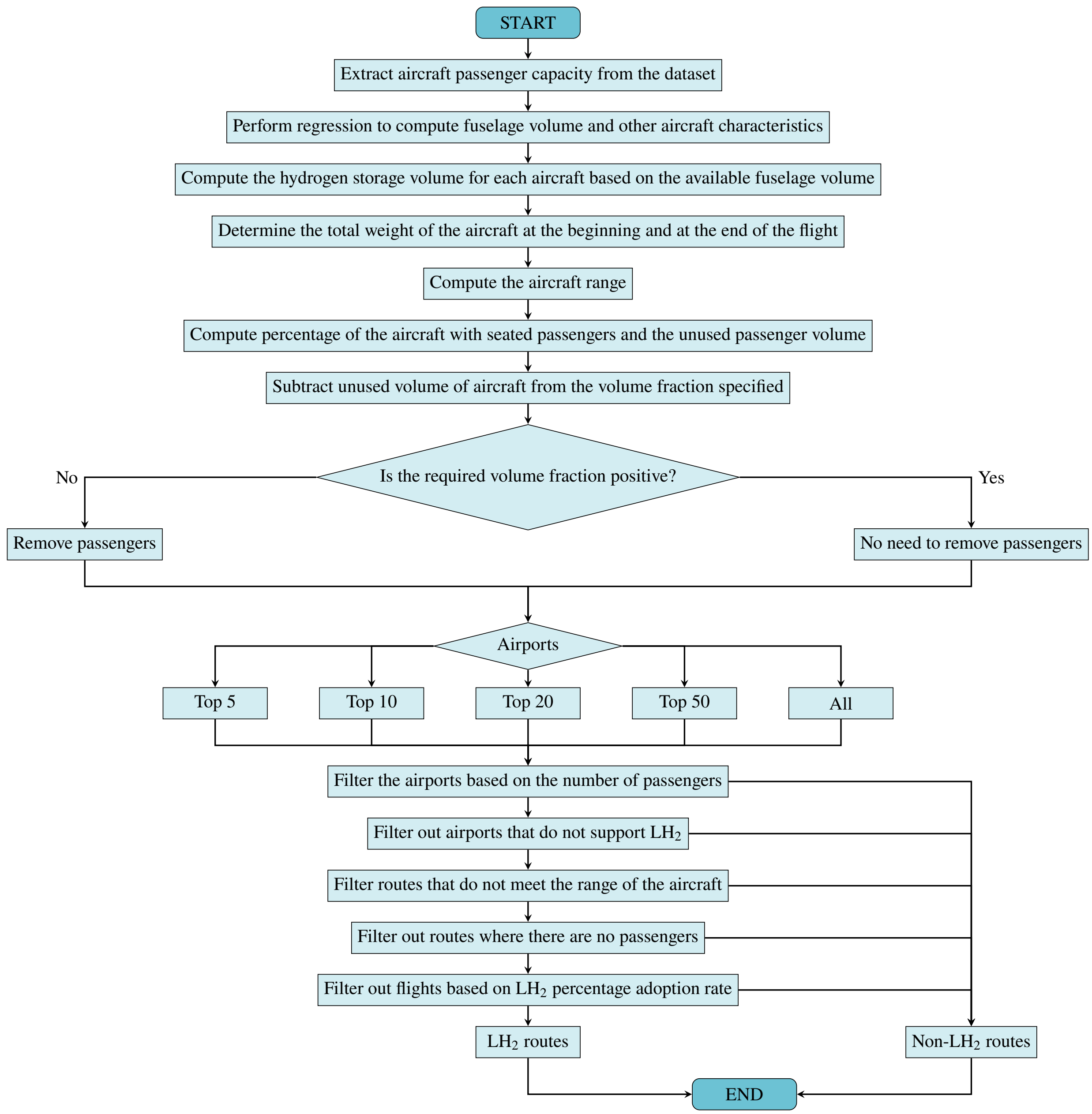

Fig. 3 Flowchart of the hydrogen route selection process implemented in the dashboard.

IV. Economic Model: Energy Source Cost Metric

The dashboard evaluates the economic implications of each energy pathway using an energy source cost metric. This metric is intended as a cost per passenger mile proxy that accounts only for the cost of the energy carrier used by the aircraft. It is not a full airline cost model. Instead, it isolates the contribution of Jet-A, electricity, SAF, or hydrogen to the route level operating cost so that different energy pathways can be compared under consistent assumptions.

For a given set of routes, the passenger mile denominator is computed as

$$PM = \sum_{r} D_r N_{\text{pax},r}, \tag{60}$$

where PM is the total passenger miles, D_r is the distance of route r, and $N_{\text{pax},r}$ is the passenger count assigned to route r. For electric and hydrogen routes, $N_{\text{pax},r}$ corresponds to the adjusted passenger count after passenger displacement is applied. Route distance, passenger count, fuel cost, flight frequency, and fuel use quantities are obtained from the flight operations dataset [1].

For conventional Jet-A operation, the energy source cost is taken directly from the route level fuel cost reported in the flight operations dataset. The corresponding energy source cost metric is

$$CASM_{E,\text{JetA}} = 100\frac{C_{\text{JetA}}}{PM_{\text{JetA}}}, \tag{61}$$

where $CASM_{E,\text{JetA}}$ is the Jet-A energy source cost in cents per passenger mile, C_{JetA} is the total Jet-A fuel cost for the selected route set, and PM_{JetA} is the corresponding passenger mile total. The factor of 100 converts dollars per passenger mile to cents per passenger mile.

For battery electric and Energy-X operation, the energy source cost is computed from the battery energy required by each feasible route, the user specified electricity price, and the number of monthly flights. The total electricity cost is

$$C_E = \sum_r E_{\text{bat},r}\left(2.77778 \times 10^{-7}\right) c_{\text{elec}} N_{\text{flights},r}, \tag{62}$$

where C_E is the total electricity cost, $E_{\text{bat},r}$ is the battery energy for route r in joules, c_{elec} is the electricity price in dollars per kilowatt hour, and $N_{\text{flights},r}$ is the number of monthly flights on route r. The conversion factor 2.77778×10^{-7} converts joules to kilowatt hours. The corresponding electric energy source cost metric is

$$CASM_{E,\text{electric}} = 100\frac{C_E}{PM_E}, \tag{63}$$

where PM_E is the passenger mile total assigned to electric operation after route feasibility, passenger displacement, airline, aircraft class, temperature, and adoption filters are applied.

For SAF operation, the dashboard computes fuel cost from the total fuel volume assigned to SAF operated routes and the user specified SAF price. The total SAF cost is

$$C_{\text{SAF}} = V_{\text{SAF,total}} c_{\text{SAF}}, \tag{64}$$

where C_{SAF} is the total SAF cost, $V_{\text{SAF,total}}$ is the total fuel volume assigned to the selected SAF scenario, and c_{SAF} is the SAF price in dollars per gallon. The SAF energy source cost metric is then

$$CASM_{E,\text{SAF}} = 100\frac{C_{\text{SAF}}}{PM_{\text{SAF}}}, \tag{65}$$

where PM_{SAF} is the passenger mile total associated with the routes assigned to SAF operation.

For hydrogen operation, the dashboard uses a user specified hydrogen cost expressed in dollars per gallon. The hydrogen storage volume is first converted from cubic meters to gallons as

$$V_{H_2,\text{gal}} = \frac{V_{H_2}}{1.0 \times 10^{-3}}\frac{1}{3.78541}, \tag{66}$$

where $V_{H_2,\text{gal}}$ is the hydrogen volume in gallons and V_{H_2} is the hydrogen volume in cubic meters. The factor 1.0×10^{-3} converts cubic meters to liters, and 3.78541 converts gallons to liters. The hydrogen fuel cost is

$$C_{H_2} = V_{H_2,\text{gal}} c_{H_2}, \tag{67}$$

where C_{H_2} is the total hydrogen cost and c_{H_2} is the hydrogen price in dollars per gallon. The hydrogen energy source cost metric is

$$CASM_{E,H_2} = 100\frac{C_{H_2}}{PM_{H_2}}, \tag{68}$$

where PM_{H_2} is the passenger mile total assigned to hydrogen operation after range, passenger feasibility, airport availability, and adoption filters are applied.

The resulting $CASM_E$ values should be interpreted as energy source cost indicators rather than complete airline cost estimates. They do not include crew, maintenance, airport charges, landing fees, aircraft acquisition, financing, insurance, infrastructure capital cost, charging or fueling equipment, storage systems, or other direct and indirect operating costs. The metric is therefore most useful for comparing the relative sensitivity of different energy pathways to fuel price, electricity price, route allocation, passenger displacement, and adoption assumptions.

V. Emissions and Life Cycle Analysis Models

The dashboard estimates monthly greenhouse gas emissions for baseline and alternative energy scenarios. These calculations are intended for screening level comparison among user defined technology pathways. The results depend on the selected emission factors, fuel volumes, fuel densities, fuel energy densities, and route allocation assumptions. Greenhouse gas emissions are reported as carbon dioxide equivalent, CO_{2e}, following a GREET style life cycle assessment framework [4].

For conventional Jet-A operation, emissions are computed from the fuel volume assigned to Jet-A routes, the density of Jet-A, and a mass based greenhouse gas emission factor. The implemented relation is

$$\mathrm{CO}_{2\mathrm{e,JetA}} = GHG_{\mathrm{JetA}} V_{\mathrm{JetA}} \rho_{\mathrm{JetA}}, \tag{69}$$

where $\mathrm{CO}_{2\mathrm{e,JetA}}$ is the Jet-A greenhouse gas emission total, GHG_{JetA} is the Jet-A emission factor in $\mathrm{kgCO}_{2\mathrm{e}}/\mathrm{kg}_{\mathrm{fuel}}$, V_{JetA} is the Jet-A fuel volume in m^3, and ρ_{JetA} is the Jet-A density in $\mathrm{kg/m}^3$. The current implementation uses

$$GHG_{\mathrm{JetA}} = 4.36466\ \mathrm{kgCO}_{2\mathrm{e}}/\mathrm{kg}_{\mathrm{fuel}}, \tag{70}$$

and

$$\rho_{\mathrm{JetA}} = 820\ \mathrm{kg/m}^3. \tag{71}$$

Fuel volumes reported in gallons are converted to cubic meters using

$$V_{\mathrm{m}^3} = 3.78541 \times 10^{-3} V_{\mathrm{gal}}, \tag{72}$$

where V_{gal} is the fuel volume in gallons and V_{m^3} is the corresponding volume in cubic meters.

For Electrification and Energy-X scenarios, the current implementation assigns zero direct operational emissions to routes classified as feasible electric routes. Routes that do not satisfy the electric feasibility filters remain assigned to conventional Jet-A operation and are included in the Jet-A emissions calculation. Upstream emissions associated with electricity generation, charging losses, battery production, and infrastructure are not included in the current implementation. Therefore, the electric emissions results should be interpreted as direct operational emissions within the dashboard boundary rather than complete life cycle emissions.

For SAF operation, emissions are computed using the life cycle emission factor and volumetric energy density of each selected fuel. The SAF life cycle emission factors and pathway assumptions are based on the SAF technology dataset and CORSIA or GREET style life cycle assessment quantities [2, 4]. For each selected fuel i, the implemented calculation is

$$\mathrm{CO}_{2\mathrm{e,SAF}} = 10^{-3} 10^{-9} \sum_i LCEF_i E_{V,i} V_i. \tag{73}$$

In Eq. (73), $LCEF_i$ is the life cycle emission factor of fuel i in $\mathrm{gCO}_{2\mathrm{e}}/\mathrm{MJ}$, $E_{V,i}$ is the volumetric energy density in MJ/L, and V_i is the volume of fuel i in liters. The factor 10^{-3} converts grams to kilograms, and the factor 10^{-9} converts kilograms to megatonnes. In the implementation, SAF volumes are converted from gallons to liters before Eq. (73) is applied.

The total emissions for a SAF scenario are computed by combining the emissions from routes assigned to SAF operation with the Jet-A emissions from routes that remain assigned to the conventional fleet:

$$\mathrm{CO}_{2\mathrm{e,with\ SAF}} = \mathrm{CO}_{2\mathrm{e,SAF}} + \mathrm{CO}_{2\mathrm{e,JetA,nonSAF}}, \tag{74}$$

where $\mathrm{CO}_{2\mathrm{e,JetA,nonSAF}}$ represents the Jet-A emissions from routes not allocated to SAF operation. The corresponding baseline assigns all routes to Jet-A and is computed as

$$\mathrm{CO}_{2\mathrm{e,without\ SAF}} = GHG_{\mathrm{JetA}} V_{\mathrm{JetA,total}} \rho_{\mathrm{JetA}}, \tag{75}$$

where $V_{\mathrm{JetA,total}}$ is the total Jet-A equivalent fuel volume for the full route set.

For hydrogen scenarios, the dashboard first computes a weighted hydrogen production emission factor when multiple production pathways are selected. If x_j is the user specified fraction assigned to pathway j, and $GHG_{H_2,j}$ is the corresponding pathway emission factor, the mixed hydrogen emission factor is

$$GHG_{H_2,\text{mix}} = \sum_j x_j GHG_{H_2,j}, \tag{76}$$

where the pathway fractions satisfy

$$\sum_j x_j = 1. \tag{77}$$

The selected hydrogen production emission factors are treated as life cycle greenhouse gas intensities for the corresponding production pathways [4]. Hydrogen emissions are then computed as

$$\mathrm{CO}_{2\mathrm{e},\mathrm{H}_2} = GHG_{H_2,\text{mix}} V_{H_2} \rho_{H_2}, \tag{78}$$

where $\mathrm{CO}_{2\mathrm{e},\mathrm{H}_2}$ is the hydrogen pathway emission total, V_{H_2} is the hydrogen volume assigned to hydrogen operated routes, and ρ_{H_2} is the liquid hydrogen density used by the dashboard.

The total emissions for a hydrogen scenario are computed as

$$\mathrm{CO}_{2\mathrm{e},\text{with } \mathrm{H}_2} = \mathrm{CO}_{2\mathrm{e},\mathrm{H}_2} + \mathrm{CO}_{2\mathrm{e},\mathrm{JetA},\mathrm{nonH}_2}, \tag{79}$$

where $\mathrm{CO}_{2\mathrm{e},\mathrm{JetA},\mathrm{nonH}_2}$ represents the emissions from routes that are not assigned to hydrogen operation. The corresponding baseline without hydrogen assigns all routes to Jet-A and is evaluated using Eq. (69) over the full route set.

The emissions outputs should therefore be interpreted within the model boundary of each module. The Jet-A and SAF calculations include fuel related life cycle emission factors as specified by the selected datasets. The hydrogen calculation reflects the selected hydrogen production pathway mix. The Electrification and Energy-X calculations currently represent direct operational emissions only and do not include upstream electricity generation or battery manufacturing emissions.

VI. Limitations

The SAT Dashboard is intended as a screening level scenario analysis tool. Its models are designed to support rapid comparison of alternative aviation energy pathways, rather than detailed aircraft design, certification analysis, or investment decision making.

The aircraft performance calculations provide first order estimates of range, energy demand, and route feasibility. Where detailed aircraft geometry or proprietary performance data are unavailable, the dashboard relies on representative aircraft parameters and regression based estimates. Therefore, the results are most useful for identifying trends and relative differences under consistent assumptions.

The Electrification and Energy-X modules use simplified range and battery pack feasibility models. These models account for battery mass fraction, specific energy, voltage, current, C-rate, and operating temperature limits, but they do not fully resolve thermal management, degradation, cell aging, transient power limits, reserve energy, pack overhead, inverter losses, motor losses beyond the user selected efficiency, or aircraft integration penalties.

The SAF module assumes that SAF can be used without aircraft range, payload, or performance penalties. This is appropriate for a drop in screening assumption, but it does not capture fuel compatibility limits, aromatics requirements, certification constraints, refinery limitations, or airport specific fuel logistics. SAF land use estimates are also based on published crop yields and feedstock conversion factors [2, 3], which may vary with agricultural practice, regional conditions, competing feedstock uses, refining capacity, policy constraints, and supply chain logistics.

The Hydrogen module uses simplified estimates of available storage volume, passenger displacement, fuel mass, aircraft weight, and range. It does not fully model cryogenic tank integration, insulation, boil off, structural reinforcement, center of gravity effects, propulsion cycle behavior, certification constraints, or airport level hydrogen production, liquefaction, storage, and distribution infrastructure.

Finally, the cost metric isolates only the energy source contribution. Other operating costs, including crew, maintenance, landing fees, aircraft acquisition, financing, insurance, and infrastructure capital costs, are not included unless explicitly added in future dashboard modules.

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